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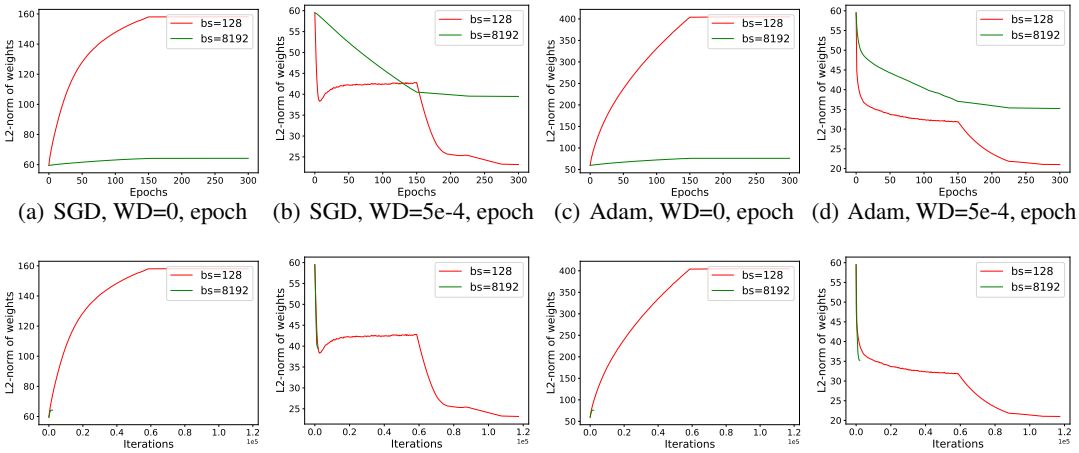
Visualizing the Loss Landscape of Neural Nets

A Comparison of Loss Surfaces

A.1 The Change of Weights Norm during Training

Figure 10 shows the change of weights norm during training in terms of epochs and iterations.



(a) SGD, WD=0, epoch (b) SGD, WD=5e-4, epoch (c) Adam, WD=0, epoch (d) Adam, WD=5e-4, epoch

(e) SGD, WD=0, iter (f) SGD, WD=5e-4, iter (g) Adam, WD=0, iter (h) Adam, WD=5e-4, iter

Figure 10: The change of weights norm during training for VGG-9. When weight decay is disabled, the weight norm grows steadily during training without constraints. When nonzero weight decay is adopted, the weight norm decreases rapidly at the beginning and becomes stable until the learning rate is decayed. Since we use a fixed number of epochs for different batch sizes, the difference in weight norm change between large-batch and small-batch training is mainly caused by the larger number of updates when a small batch is used. As shown in the second row, the changes of weight norm are at the same pace for both small and large batch training in terms of iterations.

A.2 Comparision of Normalization Methods

Here we compare several normalization methods for a given random normal direction d. Let θ_i denote the weights of layer i and $\theta_{i,j}$ represent the j-th filter in the i-th layer.

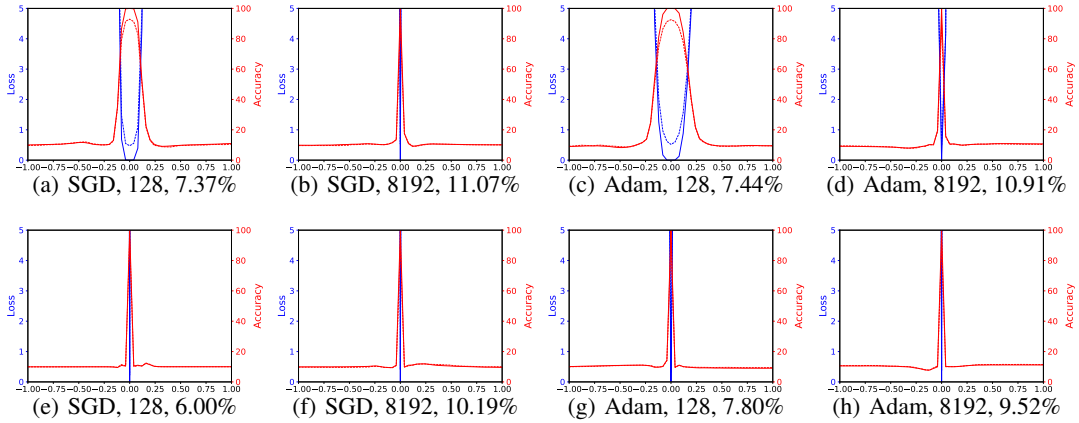
- **No Normalization** In this case, the direction d is added to the weights directly without processing.
- **Filter Normalization** The direction d is normalized so that the direction for each filter has the same norm as the corresponding filter in θ,

$$d_{i,j} \leftarrow \frac{d_{i,j}}{\|d_{i,j}\|}\|\theta_{i,j}\|.$$

  This is the approach advocated in this article, and is used extensively for plotting loss surfaces.
- **Layer Normalization** The direction d is normalized in the layer level so that the direction for each layer has the same norm as the corresponding layer of θ,

$$d_i \leftarrow \frac{d_i}{\|d_i\|}\|\theta_i\|.$$

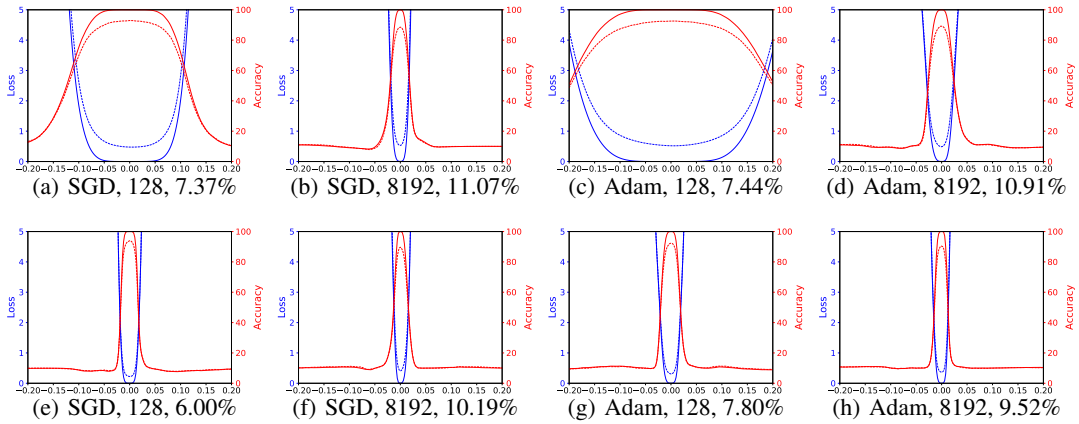
Figure 11 shows the 1D plots without normalization. One issue with the non-normalized plots is that the x-axis range must be chosen carefully. Figure 12 shows enlarged plots with $[-0.2, 0.2]$ as the range for the x-axis. Without normalization, the plots fail to show consistency between flatness and generalization error. Here we compare filter normalization with layer normalization. We find filter normalization is more accurate than layer normalization. One failing case for layer normalization is shown in Figure 13, where Figure 13(g) is flatter than Figure 13(c), but with worse generalization error.



(a) SGD, 128, 7.37% (b) SGD, 8192, 11.07% (c) Adam, 128, 7.44% (d) Adam, 8192, 10.91%

(e) SGD, 128, 6.00% (f) SGD, 8192, 10.19% (g) Adam, 128, 7.80% (h) Adam, 8192, 9.52%

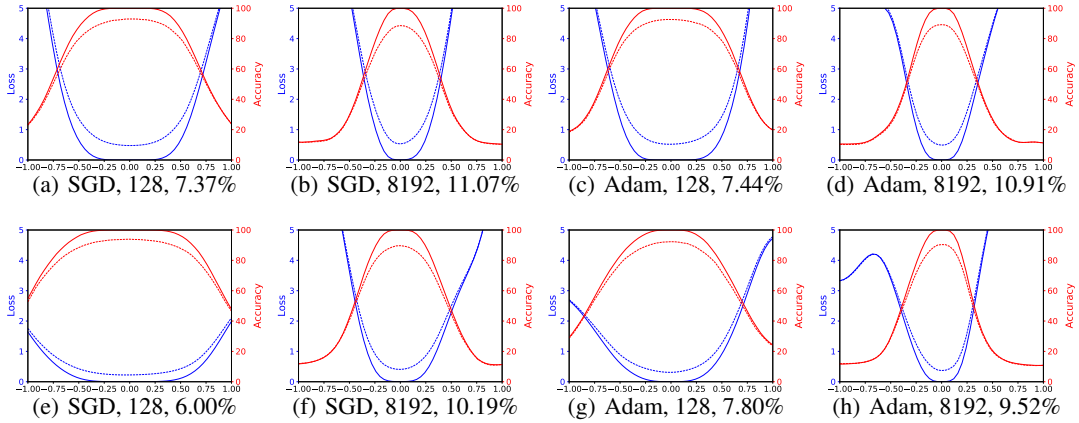
Figure 11: 1D loss plots for VGG-9 without normalization. The first row has no weight decay and the second row uses weight decay 0.0005.



(a) SGD, 128, 7.37% (b) SGD, 8192, 11.07% (c) Adam, 128, 7.44% (d) Adam, 8192, 10.91%

(e) SGD, 128, 6.00% (f) SGD, 8192, 10.19% (g) Adam, 128, 7.80% (h) Adam, 8192, 9.52%

Figure 12: Enlarged Figure 11. The range of the x-axis is [-0.2, 0.2] instead of [-1.0, 1.0. The first row has no weight decay and the second row uses weight decay 0.0005. The pairs (a, e) and (c, g) show that sharpness of minima does not correlate well with test error.



(a) SGD, 128, 7.37% (b) SGD, 8192, 11.07% (c) Adam, 128, 7.44% (d) Adam, 8192, 10.91%

(e) SGD, 128, 6.00% (f) SGD, 8192, 10.19% (g) Adam, 128, 7.80% (h) Adam, 8192, 9.52%

Figure 13: 1D loss plots for VGG-9 with layer normalization. The first row has no weight decay and the second row uses weight decay 5e-4.